Toward Generalizable Physical Attacks Against Remote Sensing Building Change Detection

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***Abstract*— Deep networks have been widely used in remote sensing building change detection (BCD) and achieved good performances. However, the BCD networks are vulnerable to adversarial attacks. Previous physical adversarial attacks in remote sensing focused on image-level tasks and instance-level tasks, and we propose a generalizable physical attack (GPA) framework for the pixel-level BCD task. The GPA framework is in the joint task-instance space to generate the universal physical adversarial patch which can cheat unseen models. It involves white-box attacks and black-box attacks. To solve the problem of the singular adversarial patch position and the singular target task in existing works, we propose an adaptive adversarial patch position module (ADPO) and present a feature loss for bitemporal remote sensing images in white-box attacks. To enhance the generalization of the patch, we propose a method in the joint task-instance space to learn the black-box adversarial patch. Extensive experiments on the widely used remote sensing image BCD dataset show that the proposed adversarial attacks against BCD achieve better performance compared with state-of-the-art methods. In particular, our GPA elevates ASR_p and CD_Rate for all buildings by an average of 29.31%, 13.92% in the white-box scenario, and 3.92%, 3.59% in the strict black-box scenario. Real-world experiments with printed patches on physical building models further demonstrate that our GPA can attack the BCD networks. The source code and associated datasets are available at https://github.com/hhhh420/adversarial_patch_BCD**

***Index Terms*— Adversarial patch, building change detection (BCD), physical attack, task-instance space.**

I. Introduction

CHANGE detection is an important task in remote sensing, especially building change detection (BCD) [1], [2], [3]. It is widely used in fields such as urban construction and coordination, and national defense security. Deep networks have demonstrated remarkable efficacy in BCD; however, these networks remain vulnerable to adversarial attacks, potentially leading to erroneous outputs despite their strong baseline performance [4], [5], [6], [7], [8]. Attackers can exploit these vulnerabilities to conceal illegal or sensitive buildings. Fig. 1 underscores the impact of physical adversarial attacks on BCD. More in line with real-world attack conditions is black-box attacks, which assume no knowledge of some or all information about the target model, as opposed to white-box attacks, which assume knowledge of the target model's parameters. However, the physical attacks in white-box and black-box scenarios for BCD pose significant challenges and require extensive research.

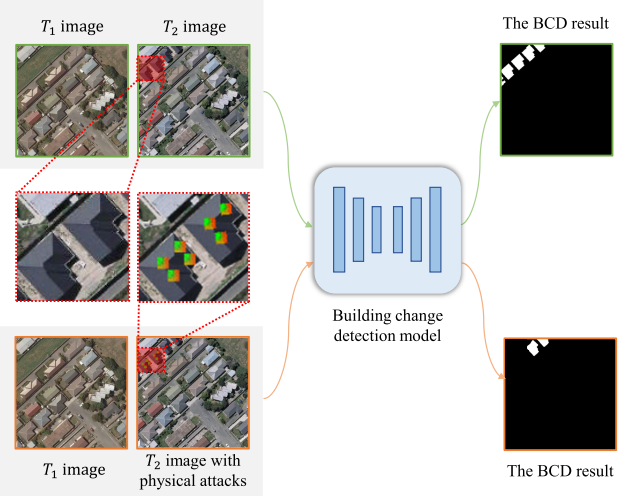


Fig. 1. Illustration of physical attacks for BCD.

This study focuses on physical attacks for BCD networks in white-box and black-box scenarios. We propose a generalizable physical attacks (GPA) framework, which involves two goals: 1) learning a universal white-box attack in physical domain to hide new buildings in BCD and 2) learning a stronger generalizable strict black-box attack which has no information including no query feedback about target models.

Goal 1 (Learning the White-Box Adversarial Patch): Although adversarial patches have proven effective in remote sensing tasks, they face certain limitations. These limitations generally manifest in two aspects.

1) Singular position of adversarial patch. Apart from the work [9], which considered the "Type Off" four-sided adversarial patch positions, most other patch positions are centered on the target. However, the existing methods are not well-suited for large objects such as buildings. Fig. 2 demonstrates their limitations.

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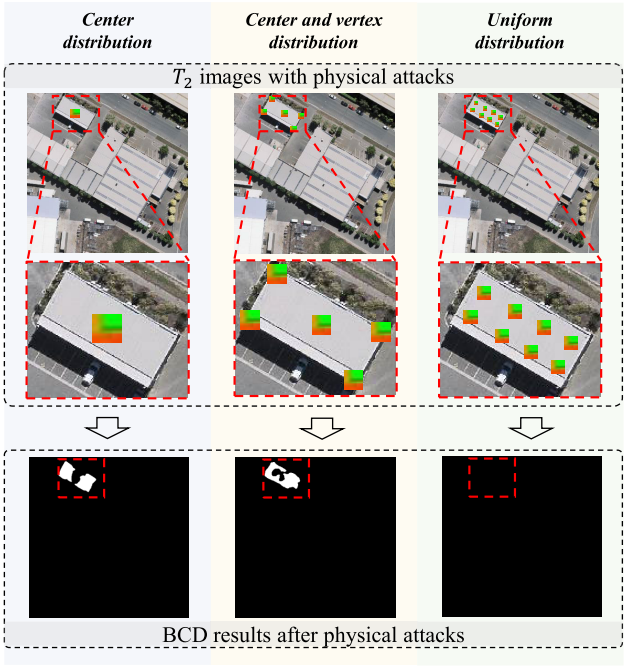


Fig. 2. Impact of patches' positions on BCD results when the area of all the patches is held constant.

2) Singular task type. In recent years, research on adversarial patches in the remote sensing field has primarily concentrated on object detection tasks, despite variations in identification categories. These instance-level methods, however, are not suitable for the pixel-level BCD task due to their lack of spatial context among pixels. Furthermore, unlike previous single-temporal image inputs, change detection tasks necessitate bitemporal images as input.

In this work, we assume that the impact of positions of patches on the results is related to the model's receptive field. The more the patches within each receptive field, the better the attack effectiveness. Fig. 2 demonstrates the varying effects of patch positions under a constant total patch area. In addition, a feature loss for bitemporal image is designed to better adapt to the BCD task.

Goal 2 (Learning the Strict Black-Box Adversarial Patch): The applicability of white-box models is curtailed in real-world scenarios due to the unavailability of the target model's structure, parameters, and query feedback. Therefore, it is crucial to devise physical adversarial attack methods tailored for strict black-box attacks to obtain more generalizable universal physical adversarial patches. Recently, numerous works have introduced meta-learning in black-box attacks to achieve better attack results [10], [11], [12], [13]. However, these methods typically require query feedback from the target model to complete adaptation [14], [15].

In this work, a physical attack in the joint task-instance space method (TIPA) is designed to improve the generalization of patches, which does not require any information about the target model and introduces meta-learning into transfer-based black-box attacks.

In summary, the main contributions of this article are as follows.

1) As far as we know, it is the first study of physical adversarial attacks on the remote sensing BCD task, revealing the vulnerability of change detection models to physical attacks.
2) We propose the GPA framework that joints task-instance space and combines the unique characteristics of the BCD task. This framework can generate adversarial patches in white-box scenarios and black-box scenarios to hide buildings.
3) We validate the effectiveness of our proposed methods and evaluate the transferability of adversarial patches across models and datasets. Furthermore, we collect a new dataset to test the generalization of physical attacks in the real world.

II. Related Work

A. Adversarial Attack on Remote Sensing

The vulnerability of deep networks [16], [17] has attracted attention in the field of remote sensing. The initial research [18] revealed the susceptibility of the existing deep networks for remote sensing scene classification to adversarial attacks. Fast-C&W [19] is an adversarial attack method for SAR target recognition. BAS [20] is a targetfool adversarial attack algorithm for image classification tasks, followed by the design of the MFCANet [21] for multimodal image classification. Furthermore, multimodal multiloss cooperative attack [22] is also an adversarial attack for multimodal image classification that combines balanced perturbation loss and cooperative adversarial loss to attack multimodal models. And MSASS [23] achieved superior attack performances on victim areas while maintaining high accuracies of other areas for semantic segmentation.

In recent years, the research of adversarial patches in remote sensing field focuses on the task of object detection. The patches are put on the aircraft and explored the effects of different patch configurations [24]. Then Patch-Noobj [25] proposes the scale-adaptive adversarial patch to adapt to the size of the object in the remote sensing image. The multiscale adversarial patch [26] is proposed and tested the effectiveness of the patch in the real world. And Sun et al. [27] propose single-color strip-based patches to hide the objects from being detected correctly in optical aerial detection. To further simulate the real world environment, "off-car" patch [9] designs the weather transformation to adapt the weather and seasonal changes.

To the best of our knowledge, previous works focused more on the image-level classification task and the instance-level object detection task, but neglected pixel-level tasks including the BCD task.

B. Physical Adversarial Attack

Adversarial attacks refer to changing input samples to make the model output incorrect results. Based on the attack domain, they can be divided into digital attacks and physical attacks.

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Algorithm 1 Physical Attack in the Joint TIPA

Input: Training Set $\mathbb{D}$, surrogate model f_i, learning rate $l_{\alpha_1}, l_{\alpha_2}, l_\beta$, hidden building mask M, total iterations T, project $\prod_\varepsilon$, loss function L, hyperparameter γ.
Output: Adversarial patch perturbation δ.

```
Randomly initialize δ;
while not done do
    for batch 𝔹 = {X_1^i, X_2^i, y^i} ~ 𝔻 do
        Sample minibatch of data 𝕊 and ℚ from 𝔹;
        for all f_i (i = 1, 2, ..., n) do
            Evaluate ∇_δ L(f_i(.), 𝔹, δ, M) using batch 𝔹;
            Evaluate ∇_δ L(f_i(.), 𝕊, δ, M) using minibatch 𝕊;
            Compute adapted perturbation via (6);
            Project δ_i' ← ∏_ε(δ_i');
            Evaluate ∇_δ L(f_i(.), ℚ, δ', M) using minibatch ℚ;
        end for
        Compute δ_task via (7);
        Compute δ_ins via (8);
        Update δ via (9);
        Project δ ← ∏_ε(δ);
    end for
end while
return δ.
```

Physical attacks are implemented in the real world and the attack form can be divided into adversarial patches, optical adversarial attack, and adversarial 3-D-print. Compared with the others, the adversarial patch is easier to achieve in the real world. In general, the TV Loss [28] is used to make adversarial patches look smoother, and the NPS Loss [29] is used to make the adversarial patch color closer to the printer color. There are numerous studies focusing on physical attacks in white-box scenarios [30], [31], [32]. In addition, CAMOU [33] learns physical vehicle disguises to perform black-box adversarial attacks. GenAP-DI [34] enhances model transferability by perturbing the adversarial sample data generator. PRAD [35] introduces the center dot to simulate the real environment changes which can fool the commercial traffic sign recognition system in the decision-based black-box setting. Meta-attack [36] is the first work to formulate physical attacks using both digital and physical images.

However, the aforementioned black-box attack works only focus on learning adversarial patches in a single task space or instance space. By combining both, not only can common features across tasks be learned but also the unique features of each task can be preserved. Therefore, this work emphasizes the learning of physical attacks in the task-instance space.

III. Methodology

We consider the universal physical adversarial attack setting where an identical attack is effective against different networks [37] and the attack can be implemented in the real world. This type of attack is easy to operate. Adversarial patches can be used to hide sensitive military buildings. This article studies physical attacks against BCD and is the first study of adversarial attacks for BCD task. Next, we formalize the physical attacks for BCD and propose white-box attacks and black-box attacks against BCD.

A. Problem Formulation

Given a pair of clean images X_1 and $X_2 \in R^{W\times H\times C}$, where W, H, and C represent the width, height, and number of channels, respectively. A network is trained to minimize the loss function

$$L(X_1, X_2) = \sum_{j=1}^{W\times H} J\left(y_j^{\text{true}}, f(X_1, X_2)_j\right) \tag{1}$$

where y_j^{true} represents the $j_$th pixel of the change label, $f(X_1, X_2)_j$ denotes the $j_$th pixel of the predicted change detection result, and J represents the loss function, which in this study is specifically implemented as the focal loss.

For the more realistic simulation against attack, only X_2 which is the posttemporal image is disturbed by the patch, and the pretemporal image X_1 remains original. Because in the real-world we can only change the present, not the past. Our goal is to add an adversarial patch $\delta \in R^{W'\times H'\times 3}$ to the perturbation mask M, causing the results to be erroneous. Therefore, we denote the perturbation mask $M \in \{0, 1\}^{W\times H}$ where $M_j = 1$ indicates that the $j_$th pixel can place the patch, otherwise $M_j = 0$. Let $F \in \{0, 1\}^{W\times H}$ denotes the hidden building mask such that $F_j = 1$ means the $j_$th pixel should be misclassified by the attacker, and $F_j = 0$ means the $j_$th pixel needs to be classified (including as changed or unchanged) correctly. And set y^{pred} is the predicted result of the clean samples X_1, X_2, and the BCD against physical attacks can be expressed as

$$\begin{aligned}\delta^* = \arg\min & \sum_{j|F_j=1} -J\left(y_j^{\text{pred}}, f(X_1, X_2 + M\odot\delta)_j\right) \\ & + \sum_{j|F_j=0} J\left(y_j^{\text{pred}}, f(X_1, X_2 + M\odot\delta)_j\right).\end{aligned} \tag{2}$$

B. Learning to Generate White-Box Physical Patches

For the training to generate physical adversarial patches, we adopt a classic network structure [38]. To better align it with the BCD task, we propose the adaptive adversarial patch position module (ADPO) and a loss function targeting the feature level of bitemporal images. The pipeline of the white-box attack with uniform distribution position and feature consistency constraint (FPW) is shown in Fig. 3.

1) Patch Position and Shape: In general, for small targets (e.g., traffic signs), adversarial patches cover the entire target or random position [39], while for large targets (e.g., airplanes), adversarial patches are usually placed at the center of the target [25], [26]. The buildings we focus on belong to large targets. From Fig. 2, it can be observed that although all types of adversarial patches have the same total area, the more spread out the patches, the better the attack works. And each patch only affects a limited area around it. Therefore, we hypothesize that placing patches more uniformly would better hide the building, due to the receptive field constraints.

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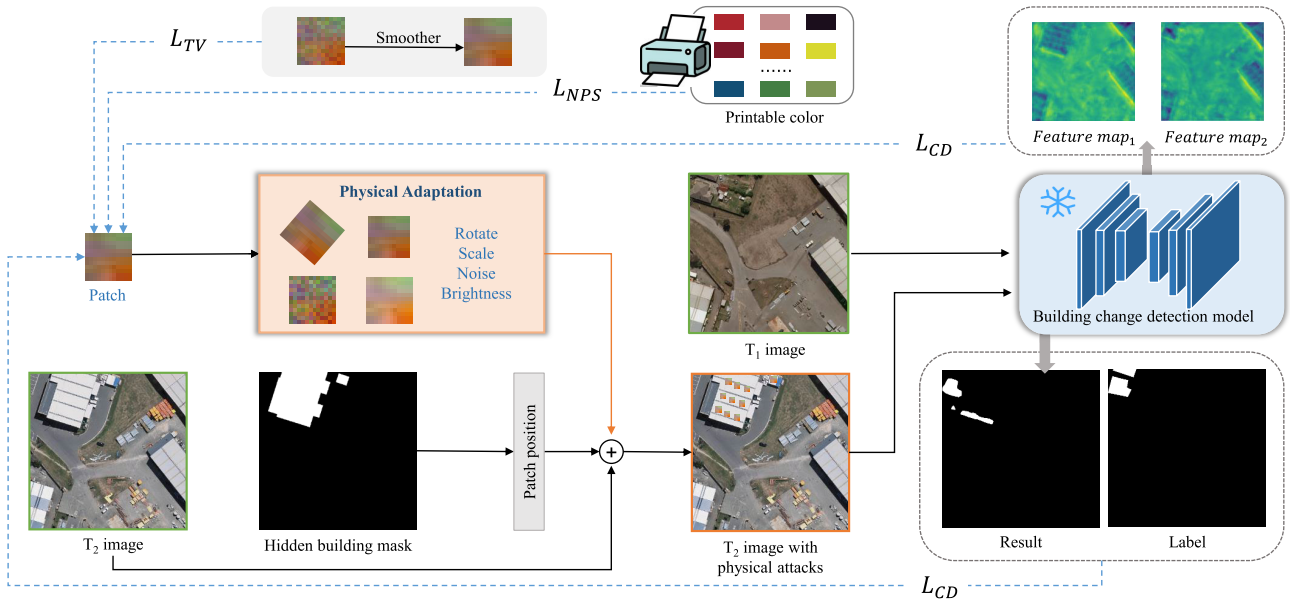


Fig. 3. Flowchart of the proposed FPW white-box physical attack method. The "hidden building mask" refers to the buildings that are intended to be hidden, namely, those that exist commonly in both the T_2 label and the change label. And the "label" is acquired through subtracting the hidden building mask from the change label.

We propose the ADPO module to achieve a uniform distribution of patches on the building. The flow of the entire module is as follows.

1) Calculate the number of patches that can be placed in the length and width of each building according to the preset ratio of patches area to building area.
2) $y = kx + b$ is used to represent the equations of lines parallel to the length and width of the smallest external rectangle of the building, respectively. The intersections of these lines are calculated as the center positions of the patches.
3) Ensure that the patch is positioned on the building. If the patch deviates from the building's surface and the building is designated to receive only one patch, place the patch at the building's center. If, however, multiple patches are considered but some deviate from the building, only place those patches that have at least one point of the building within their radius (defined by the patch size). Delete any patches that do not meet this criterion.

For the patch shape, we choose the square due to their ease of handling and minimal susceptibility to orientation. Unlike previous works [26], [38], this study adopts a fixed patch size that is independent of the building's size and shape, ensuring both universality and operational simplicity. Since all the images are taken from an overhead perspective, the influence of building height, compared with the image acquisition height, can be ignored.

2) Loss Function: We adopt the widely used TV Loss [28] denoted L_{TV} and NPS Loss [29] denoted L_{NPS} from generative adversarial patches to ensure the smoothness of the patch and their color similarity to the printer.

Loss functions are crucial for enhancing the perturbations of the patch. In this study, we propose using L_{CD} at two levels. The first level concerns the BCD result. We use the focal loss L_{fl} [44] to address the class imbalance issue

$$L_{\mathrm{fl}} = \begin{cases} -m(1-p)^n \log(p), & y^{\mathrm{true}} = 1 \\ -(1-m)p^n \log(1-p), & y^{\mathrm{true}} = 0 \end{cases} \tag{3}$$

where p stands for the estimated probability. And according to [44] we adopt $m = 0.75$ and $n = 2$. The second level focuses on features. The feature maps corresponding to the changing areas in the bitemporal images exhibit significant disparities. By making the feature maps of the region to be concealed in the posttemporal images (denoted as T_2) more similar to those in the pretemporal images (denoted as T_1), the objective of concealing buildings can be achieved. Consequently, we calculate the Euclidean distance between the feature maps of the original T_1 image and the adversarial T_2 image

$$L_{\mathrm{fe}} = \sum_{i \in F} \left\| X^2_{\mathrm{feature}_i}{}' - X^1_{\mathrm{feature}_i} \right\|_2^2 \tag{4}$$

where F is the location where the pixel value is 1 on the hidden building mask. Consequently, the total loss function is defined as

$$\mathrm{Loss} = L_{\mathrm{CD}} + \alpha_1 L_{\mathrm{TV}} + \alpha_2 L_{\mathrm{NPS}} \tag{5}$$

where $L_{\mathrm{CD}} = L_{\mathrm{fl}} + \beta L_{\mathrm{fe}}$. The term L_{fl} is defined in (3), while L_{fe} is defined in (4).

C. Learning to Generate Black-Box Physical Patches

To enhance the generalization of patches, a strict black-box attack TIPA method is proposed as shown in Fig. 4, where the target model is hidden from the attacker and queries are not allowed. Meta-learning also known as "learning to learn," that is, the use of previous knowledge and experience to guide the learning of new tasks, so that the network has the

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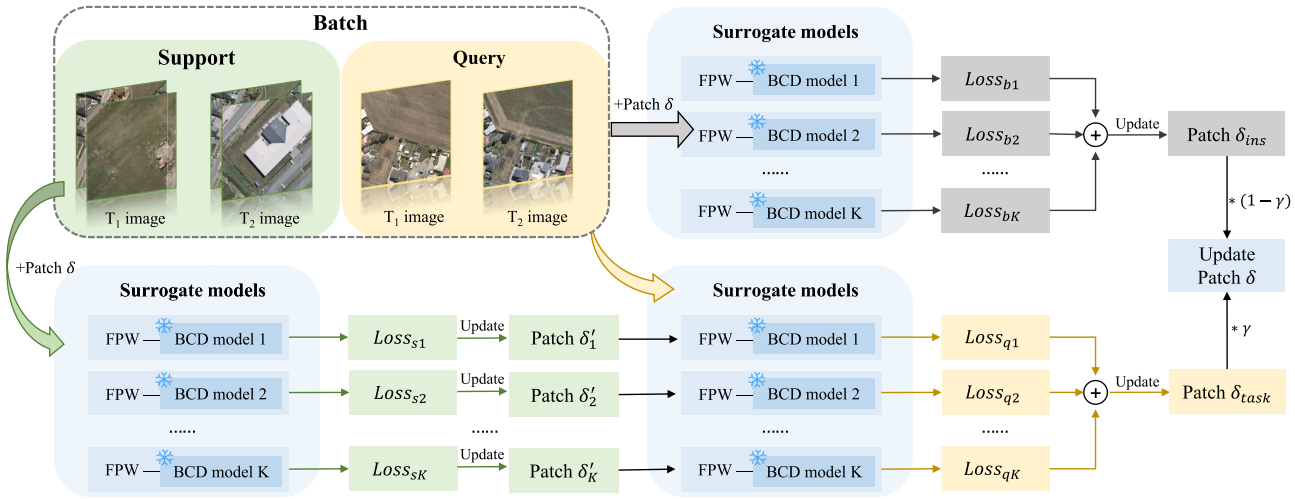


Fig. 4. Flowchart of the proposed TIPA black-box physical attack method. A batch is used for training the instance-level patch perturbation δ_{ins}, and within this batch, samples are randomly selected to serve as support and query sets for training the task-level patch perturbation δ_{task}. Surrogate models refer to BCD models whose information are available, and they surrogate the target models in the training stage to train patches.

ability to learn [45], [46], [47]. The trainable network's part includes initialize parameters, optimizer, network architecture search, data augmentation, and sample weighting. The most classical model is MAML [45], which trains initialization parameters in the task space and then completes the adaptation on the target task. In this work, we take obtaining generalized adversarial patches as the training initialization parameters in meta-learning. Although there is no information and feedback from the target model in the strict black-box scenario, the use of meta-learning according to [37], [48] is expected to achieve the goal of generating generalized patches without adaptation.

1) Task Definition: Task is the atom in meta-learning. In the strict black-box scenario, task is defined as "given a set of clean bitemporal images and a target model, the goal is to learn an adversarial patch that can successfully against the target model." In the meta-train task, the training set and the test set are referred to as the support set and the query set. And the BCD model for parameters and structures acts as a surrogate for the target model in the training stage (i.e., surrogate model).

2) Task-Instance Space: Meta-learning is trained in the task space and lacks information in the instance space. In the strict black-box scenario, transfer learning, ensemble learning, and other methods in the instance space are more used at present. In addition, only in the task space training has poor effect when the number of samples increases [52]. So we propose a TIPA framework that incorporates both the task space and instance space in updating parameters. The overall training process is listed in Algorithm 1.

In the task space, given a support set $\mathbb{D}_s$ and a query set $\mathbb{D}_q$, we follow the MAML [45] to perform meta-learning. The inner loop optimization on the support set can be represented as

$$\delta_i' = \delta - l_{\alpha_1} \cdot \nabla_\delta L(f_i(.), \mathbb{D}_s, \delta, M) \tag{6}$$

where $f_i(.)$ is the i_th surrogate model, δ_i is the pixels of the adversarial patch computed by the i_th surrogate model, and M is the hidden building mask. L denotes the loss function of white-box adversarial patch training, as previously introduced in (5). Then, for optimizing the outer loop on the query set, given n surrogate models, the pixels of the patch are updated as

$$\delta_{\text{task}} \leftarrow \delta - l_{\alpha_2} \cdot \frac{1}{n} \sum_{i=1}^{n} \nabla_\delta L\left(f_i(.), \mathbb{D}_q, \delta_i', M\right). \tag{7}$$

In the instance space, given a training set $\mathbb{D}$, we take the bitemporal images as the atom. The losses for each bitemporal image across all the surrogate models are summed, followed by parameter updates

$$\delta_{\text{ins}} \leftarrow \delta - l_\beta \cdot \frac{1}{n} \sum_{i=1}^{n} \nabla_\delta L(f_i(.), \mathbb{D}, \delta, M). \tag{8}$$

Therefore, in the joint task-instance space, the parameter updates are as follows:

$$\delta \leftarrow \gamma * \delta_{\text{task}} + (1 - \gamma) * \delta_{\text{ins}}. \tag{9}$$

IV. Experimental Results

A. Datasets and Evaluation Metrics

1) Datasets: To simulate real-world operations, the dataset needs to have known spatial resolution. Therefore, digital experiments are conducted on the widely used BCD datasets WHU Building Change Detection dataset (WHU-BCD) [53] and LEVIR-CD [54]. WHU-BCD comprises a pair of aerial images of the same area, with a resolution of 32 507 × 15 345 and a spatial resolution of 0.3 m, including RGB channels. As no data split solution is provided in [53], we crop the image into small patches of size 512 × 512 with no overlap and randomly split it into training set, validation set, and test set with 1344, 192, and 384 images, respectively. LEVIR-CD consists of 637 image patch pairs with a size of 1024 × 1024 pixels and a spatial resolution of 0.5 m. According to [42], we crop each image into nonoverlapping patches with size of 256 × 256. The dataset randomly applies 7120, 1024,

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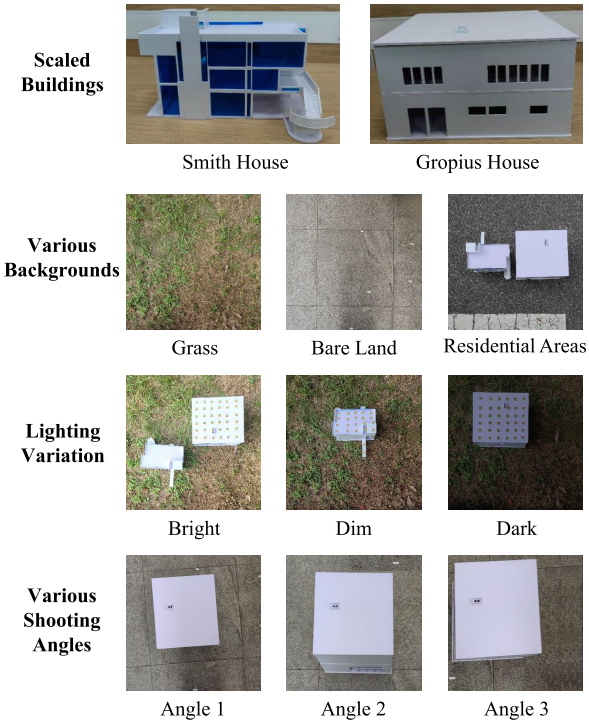


Fig. 5. Rationality of physical experimental setups.

and 2048 pairs of patches for the training set, validation set, and test set, respectively.

For physical experiments, due to the limitations of real conditions and funding, it is difficult to perform large-scale patch adhesion and image acquisition on real buildings. According to [55], [56], we conduct scale-down experiments. As shown in Fig. 5, miniature buildings and patches scaled down by 1:200 are used for scaled experiments. To enhance the diversity of building types, miniature models of real-world structures, such as Smith House and Gropius House, are selected and used. In addition, various scenarios are chosen to mimic bare land, grass, and residential areas, increasing augmenting the diversity of scenes represented in our dataset. To capture a range of lighting conditions, images were collected under different intensities of illumination, specifically during the summer season in Wuhan, China, at 12:00–13:00 (representing bright), 17:00–18:00 (representing dim), and 20:00–21:00 (representing dark). Furthermore, images are captured from multiple viewpoints, including directly overhead (top–down view of rooftops), oblique angles of rooftops, and oblique angles incorporating the facades of buildings, to ensure that our dataset closely approximates full-scale real-world conditions. Finally, patches measuring 0.018 × 0.018 m are printed for the physical experiment based on a 1:200 scale. A total of 100 sets of data are collected, with each set of data containing a pretemporal image (T_1 image), a posttemporal image (T_2 image), a T_2 image with physical attacks, and a label indicating the change between T_1 image and T_2 image (change label). And the pixel-level annotation is performed using LabelMe. Data samples are shown in Fig. 6. The acquisition conditions for T_2 image and T_2 image with physical attacks are not entirely identical, which better aligns with real-world scenarios and demonstrates the adaptability of adversarial patches to various environments. The entire dataset is used to test the generalization of the patches and is not used in the training stage.

2) Evaluation Metrics: The attack performance is quantitatively evaluated using the widely adopted metric attack success rate (ASR) [57]. The rates of successful attacks of pixels in the adversarial and nonadversarial regions are denoted by ASR_p and ASR_t, respectively. Following [58], [59], [60], we also report the mean intersection over union (mIoU) and $F1$-score ($F1$) to evaluate the performance. In addition, to determine whether buildings are completely hidden, we design a new evaluation metric complete destroy rate (CD_Rate). This metric represents the proportion of successfully deceived models that completely hide buildings out of the total number of buildings that need to be hidden. To better evaluate the patches effectiveness, buildings are categorized based on their area. Specifically, buildings with an area less than 800 square meters were classified as small, those exceeding 3000 square meters as large, and the rest as medium structures.

B. Experiments of White-Box Digital Attacks

1) Experimental Details: In the white-box scenario, we conduct experiments only on WHU-BCD. The BCD models are trained on the clean original training set of WHU-BCD, and during white-box attack training, the parameters are frozen and not involved in the training and testing process. We simulate real-world operations by adding patches to the posttemporal images. Because we cannot alter the before, only the present in reality. And the roof's inclination angle is disregarded. We adopt Adam [61], [62] and the learning rate is set to 0.0001. The training epoch is set to 100. According to [9], the hyperparameters α_1, α_2 are set to 2.5 and 0.01, respectively. And the hyperparameter β are set to 0.005. The patch-to-building area ratio is 10%, with the patch size being 12 × 12 pixels (i.e., 3.6 × 3.6 meters). We use a single NVIDIA RTX 3090 GPU for both training and testing.

2) Competing Methods: No physical adversarial attack targeting BCD specifically has been identified in the current literature. To validate the effectiveness of training with patches, a random noise generated image which denoted noise and a randomly selected remote sensing image which denoted RS img are used as comparison patches. The patches positions, shapes, and sizes are kept consistent in settings.

3) Experimental Results: To validate the effectiveness of adversarial patches, experiments are conducted on three BCD networks, with the results shown in Table I. The bold font indicates the optimal results. Our primary focus is on the results in the attacked regions, with ASR_t serving as an evaluation of the nonattacked areas, albeit with a magnitude difference of 10^{-3}, hence it is not given in bold. BIT [40], ISNet [41], and DMINet [42] (with ResNet34 [63] adopted as the backbone) achieved results of 89.73%, 89.94%, and 88.30% on the clean original test set of mIoU, respectively. In addition, their corresponding $F1$-scores were recorded as 89.03%, 89.28%, and 87.31%. And the CF-GCN [43]

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Fig. 6. Samples of physical dataset. Each data sample includes a pretemporal image (T_1 image), a posttemporal image (T_2 image), a T_2 image with physical attacks, and a label indicating the change between T_1 image and T_2 image (change label).

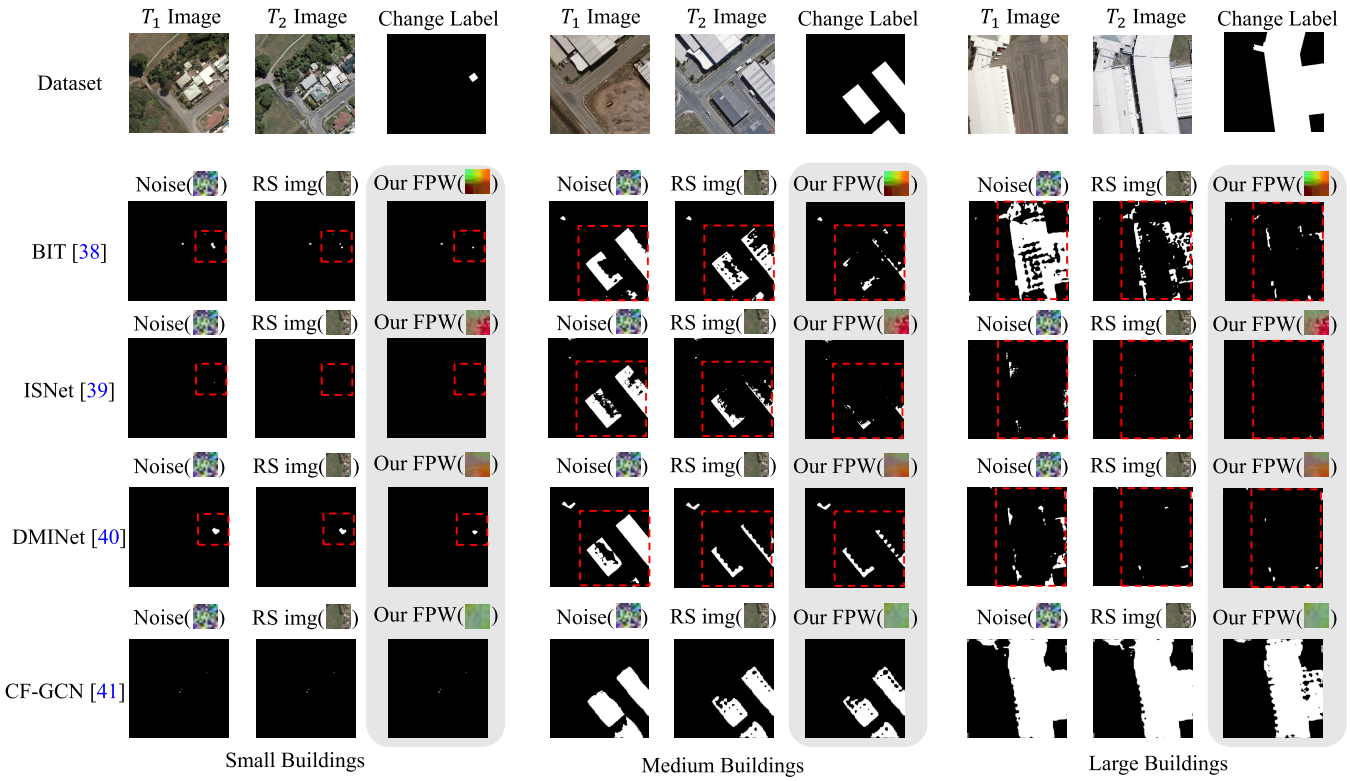
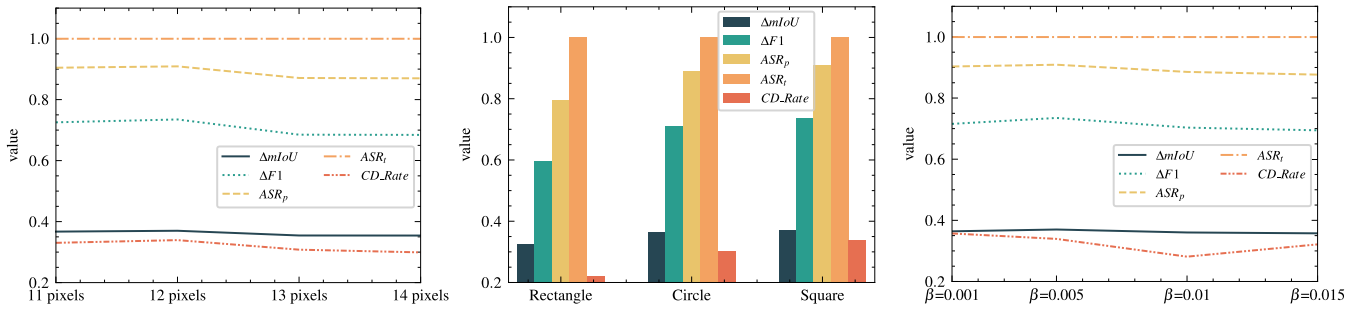


Fig. 7. White-box visualization results on different BCD networks. BIT [40], ISNet [41], DMINet [42], and CF-GCN [43] are the BCD networks. Noise, RS img, and our FPW are the attack methods.

method, which is based on graph convolutional network (GCN), achieved the mIoU of 84.39% and $F1$ of 82.35%. The ΔmIoU and $\Delta F1$ represent the decrease in mIoU and $F1$ compared with adversarial results and original results. Our method demonstrates improvements of at least 7.12% in all the evaluation metrics. In particular, the ASR_p metric showed 40.57% improvement on the BIT model. It also yielded favorable outcomes in segmented categories. Fig. 7 presents the visualized results and trained adversarial patches of our method on the test set.

The ablation study is conducted on the BIT. When without using the ADPO module, patches are placed at the center of the minimum bounding rectangle of the building. The experimental results in Table II demonstrate that both our proposed ADPO module and the feature loss for bitemporal images contribute to improve the performance of attacks.

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(a) (b) (c)

Fig. 8. Parameter analysis of the white-box FPW method on BIT model. (a) Parameter analysis of patch size. (b) Parameter analysis of patch shape. (c) Parameter analysis of β.

TABLE I
COMPARISON TO OTHER WHITE-BOX ADVERSARIAL PATCH METHODS

BCD Networks	Attack Methods	All					Small		Medium		Large	
		$\Delta mIoU$	$\Delta F1$	ASR_p	ASR_t	CD_Rate	ASR_p	CD_Rate	ASR_p	CD_Rate	ASR_p	CD_Rate
BIT [40]	Noise	10.66	14.27	25.76	99.66	11.13	30.57	12.02	22.46	0.00	13.66	0.00
	RS img	17.20	25.00	43.09	99.66	13.17	31.97	14.03	56.14	3.12	66.23	0.00
	Our FPW	**32.92**	**60.31**	**83.66**	99.66	**28.39**	**82.67**	**30.26**	**79.47**	**6.25**	**90.47**	0.00
ISNet [41]	Noise	26.71	43.91	48.88	99.94	13.36	37.54	14.23	58.44	3.12	77.55	0.00
	RS img	26.26	43.13	59.50	99.95	16.14	46.79	17.03	72.54	3.12	89.34	**12.50**
	Our FPW	**35.56**	**67.86**	**87.80**	99.92	**31.91**	**87.58**	**33.47**	**81.62**	**12.50**	**94.73**	**12.50**
DMINet [42]	Noise	24.06	39.87	42.39	99.97	11.69	36.36	12.63	37.88	0.00	67.30	0.00
	RS img	24.15	40.08	64.13	99.96	16.51	52.69	17.64	75.11	3.12	91.96	0.00
	Our FPW	**31.27**	**58.01**	**83.20**	99.65	**27.27**	**79.95**	**29.06**	**82.41**	**6.25**	**94.04**	0.00
CF-GCN [43]	Noise	2.02	2.75	7.88	99.89	16.14	3.57	17.03	13.56	6.25	16.43	0.00
	RS img	7.85	11.51	24.82	99.81	22.08	20.53	23.25	36.80	**9.38**	27.52	0.00
	Our FPW	**10.55**	**15.99**	**32.30**	99.81	**22.82**	**30.55**	**24.05**	**38.31**	**9.38**	**32.40**	0.00

TABLE II
ABLATION STUDY ON THE PROPOSED WHITE-BOX ATTACK METHOD

ADPO Module	Feature Loss	All					Small		Medium		Large	
		$\Delta mIoU$	$\Delta F1$	ASR_p	ASR_t	CD_Rate	ASR_p	CD_Rate	ASR_p	CD_Rate	ASR_p	CD_Rate
×	×	16.03	22.95	40.03	99.96	13.91	52.00	15.03	25.49	0.00	15.59	0.00
✓	×	30.36	53.37	77.01	99.96	20.59	76.07	22.04	73.35	3.12	83.21	0.00
✓	✓	**32.92**	**60.31**	**83.66**	99.96	**28.39**	**82.67**	**30.26**	**79.47**	**6.25**	**90.47**	0.00

However, the performance of CD_Rate in the case of large buildings is poor, even reaching 0.00%. This is attributed to the adoption of the most stringent evaluation criteria by this metric, which necessitates that every pixel of the building intended for concealment be recognized as an unchanged area. Small buildings, medium buildings, and large buildings are classified based on their area, specifically the number of pixels they cover. This characteristic facilitates the complete concealment of small buildings, thereby yielding better CD_Rate performance. Conversely, for large buildings, even if only a few pixels remain unhidden, the CD_Rate performance may suffer significantly.

An analysis is conducted on the patch shape, patch size, and the hyperparameter β in the loss function L_{CD} on the validation set of WHU-BCD. Fig. 8(a) presents the experimental results of different patch sizes on BIT. To avoid difficulties in acquiring adversarial patches at lower resolutions and to prevent patches from being too large to be practically manipulated in real-world scenarios, the analysis is conducted within the range of 10 × 10 pixels (i.e., 3 × 3 m)–15 × 15 pixels (i.e., 4.5 × 4.5 m). The results indicate that the optimal performance occurs at 12 × 12 pixels, though overall, the performance is not highly sensitive to this parameter. Fig. 8(b) displays the experimental results for different patch shapes on BIT, with commonly used rectangles, circles, and squares being selected for experimentation. The results show that the square shape yields the best performance. Fig. 8(c) demonstrates the parameter analysis of β, revealing that this

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TABLE III
COMPARISON TO OTHER BLACK-BOX ADVERSARIAL METHODS ON THE WHU-BCD DATASET

BCD Networks	Attack Methods	All					Small		Medium		Large	
		$\Delta mIoU$	$\Delta F1$	ASR_p	ASR_t	CD_Rate	ASR_p	CD_Rate	ASR_p	CD_Rate	ASR_p	CD_Rate
FCCDN [51]	EA [49]	**37.91**	**72.41**	92.22	99.90	78.48	94.30	82.36	81.17	34.38	97.09	12.50
	MTA [13]	36.13	66.88	90.86	99.90	78.67	94.23	82.63	80.42	32.29	90.14	16.67
	LRS [50]	6.09	7.50	15.26	99.94	15.03	20.74	16.03	8.09	0.00	2.96	12.50
	Our TIPA	37.63	71.44	**93.14**	99.89	**82.93**	**94.66**	**86.17**	**83.43**	**46.88**	**98.56**	**25.00**
ISNet [41]	EA [49]	35.48	67.61	87.67	99.92	35.25	88.05	37.47	80.45	9.38	93.60	0.00
	MTA [13]	34.56	64.81	88.13	99.86	36.18	89.20	35.41	80.67	6.25	92.02	**4.17**
	LRS [50]	17.58	25.87	43.85	99.95	13.36	30.79	14.36	57.32	1.04	74.44	0.00
	Our TIPA	**36.36**	**70.24**	**91.07**	99.84	**41.37**	**92.12**	**43.89**	**83.57**	**12.50**	**95.04**	0.00
DMINet [42]	EA [49]	32.60	61.81	85.39	99.96	37.66	84.63	40.28	80.48	6.25	92.84	0.00
	MTA [13]	31.57	58.98	82.81	99.95	31.48	81.90	33.60	77.30	6.25	91.35	0.00
	LRS [50]	19.44	30.23	51.50	99.97	11.50	36.73	12.29	65.72	2.08	87.38	0.00
	Our TIPA	**33.90**	**65.73**	**88.91**	99.95	**41.00**	**88.45**	**43.69**	**83.76**	**9.38**	**95.57**	0.00
CF-GCN [43]	EA [49]	22.68	40.83	66.40	99.75	34.88	73.92	37.07	62.41	9.38	45.95	0.00
	MTA [13]	21.43	40.61	62.07	99.71	37.48	64.44	39.95	59.56	8.33	56.83	0.00
	LRS [50]	4.47	6.30	14.84	99.84	18.74	10.19	19.84	24.82	6.25	20.48	0.00
	Our TIPA	**27.30**	**52.89**	**78.91**	99.73	**42.12**	**85.50**	**44.89**	**69.89**	**9.38**	**66.22**	0.00

TABLE IV
CROSS DATASET COMPARISON TO OTHER BLACK-BOX ADVERSARIAL METHODS ON THE LEVIR-CD DATASET

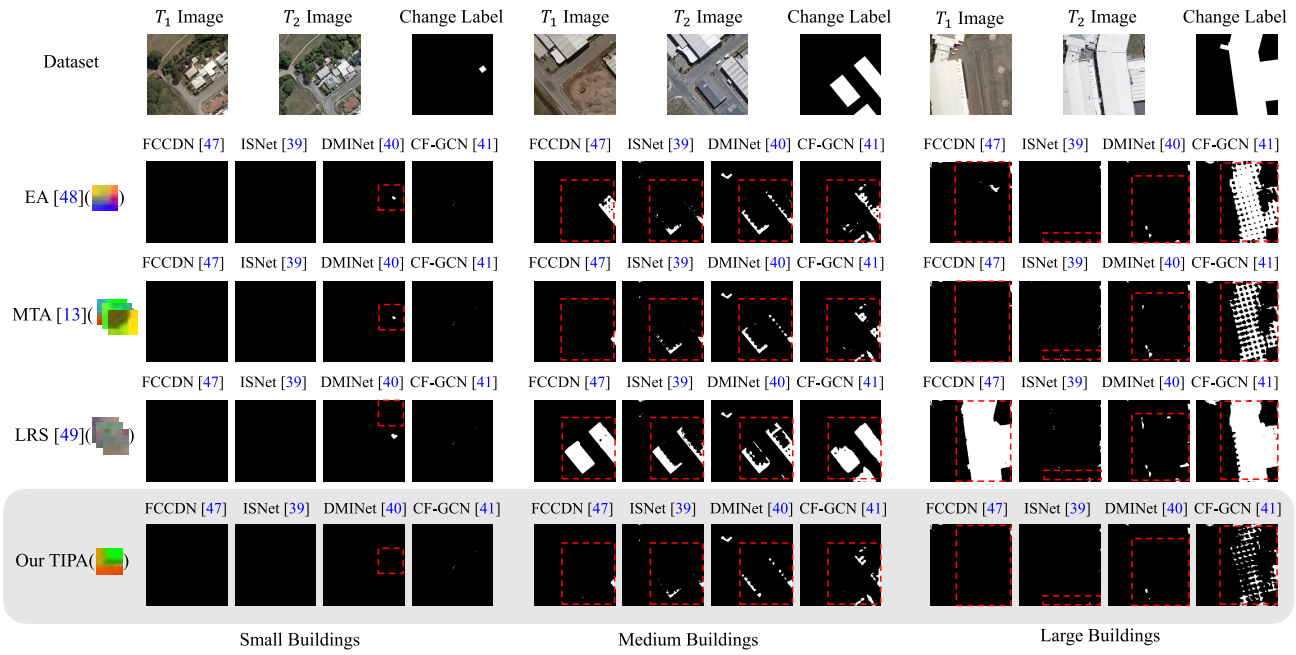
BCD Networks	Attack Methods	All					Small		Medium		Large	
		$\Delta mIoU$	$\Delta F1$	ASR_p	ASR_t	CD_Rate	ASR_p	CD_Rate	ASR_p	CD_Rate	ASR_p	CD_Rate
FCCDN [51]	EA [49]	15.19	22.48	36.81	99.73	23.29	37.73	23.21	32.37	30.16	23.69	33.33
	MTA [13]	**21.49**	**34.80**	82.57	99.56	53.98	82.45	53.95	83.27	54.50	93.37	64.82
	LRS [50]	2.93	3.76	10.13	99.82	10.18	10.57	10.09	8.77	17.99	0.37	22.22
	Our TIPA	15.97	23.87	**83.95**	99.54	**54.73**	**83.65**	**54.66**	**85.03**	**57.14**	**99.20**	**77.78**
ISNet [41]	EA [49]	21.84	36.21	38.88	99.65	15.44	40.83	15.44	31.32	7.94	12.73	0.00
	MTA [13]	27.91	45.88	67.14	99.61	38.21	70.50	38.51	48.09	12.17	**27.27**	5.56
	LRS [50]	1.81	2.13	5.99	99.82	2.49	6.19	2.47	4.92	4.76	4.00	0.00
	Our TIPA	**31.78**	**55.42**	**76.43**	99.61	**43.28**	**80.74**	**43.60**	**49.17**	**14.29**	27.24	**11.11**
DMINet [42]	EA [49]	23.51	30.66	53.16	99.71	34.68	55.33	34.99	38.77	7.94	33.94	0.00
	MTA [13]	33.26	59.30	78.96	99.58	56.68	81.01	57.19	67.75	10.58	66.39	**7.41**
	LRS [50]	3.95	4.77	11.06	99.84	6.38	11.19	6.41	7.40	4.23	8.85	0.00
	Our TIPA	**35.81**	**66.03**	**84.40**	99.56	**59.43**	**85.99**	**59.95**	**69.28**	**12.70**	**80.79**	5.56
CF-GCN [43]	EA [49]	26.33	42.55	65.37	99.47	44.63	66.79	44.88	60.30	22.22	48.81	**16.67**
	MTA [13]	29.69	53.29	72.53	99.39	54.91	74.59	55.25	67.19	28.04	45.97	11.11
	LRS [50]	3.12	3.76	10.09	99.63	3.40	10.06	3.97	11.07	5.29	10.80	7.41
	Our TIPA	**36.69**	**69.89**	**88.01**	99.41	**77.16**	**91.12**	**77.72**	**76.37**	**30.16**	**50.32**	11.11

parameter is not sensitive. Considering various evaluation metrics, $\beta = 0.005$ is chosen.

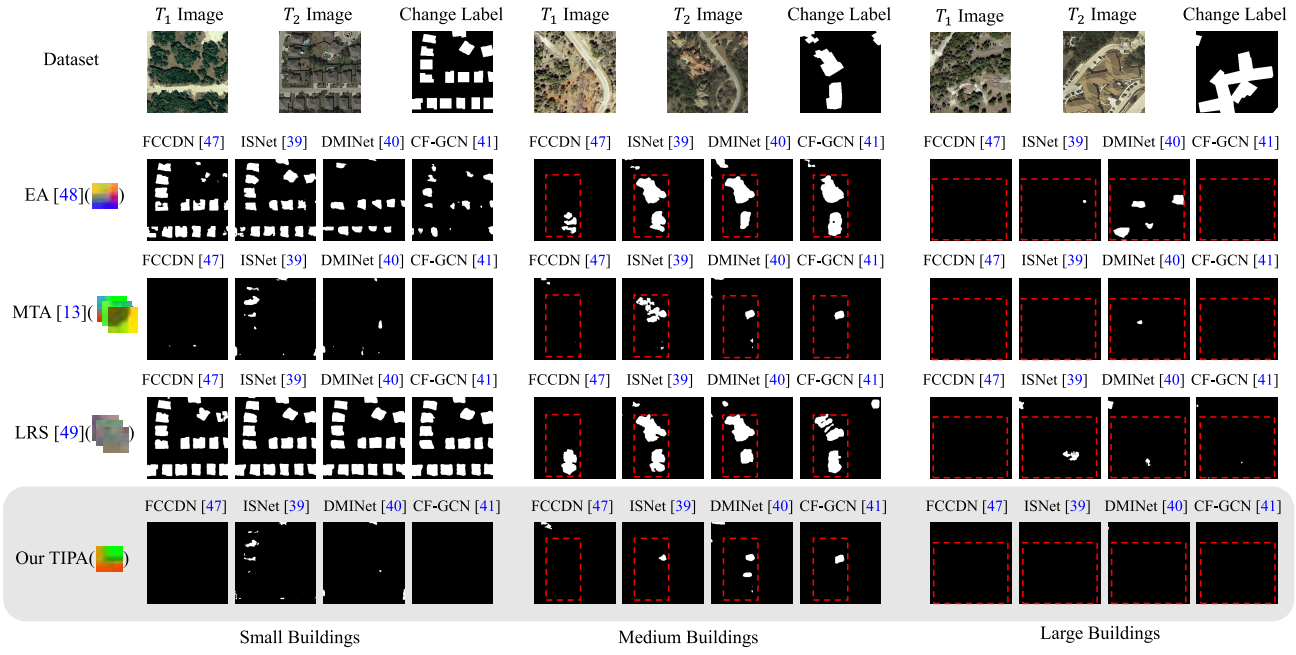
C. Experiments of Black-Box Digital Attacks

1) Experimental Details: To study the transferability of our proposed method in an open-set setting, we evaluate adversarial examples transferred across both the models and datasets. We only train on the training set of WHU-BCD and test on the both the test sets of WHU-BCD and LEVIR-CD. We randomly divide the training dataset into support set and query set with 896 and 448, respectively. For the inner loop and outer loop in the task space, the learning rate is set to 0.05 and 0.001. And the learning rate is set to 0.001 in the instance space. The hyperparameter γ in (9) is set to 0.7. The patch-to-building area ratio is 10%, with the patch size

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(a)



(b)

Fig. 9. Black-box visualization results on different BCD networks. FCCDN [51], ISNet [41], DMINet [42], and CF-GCN [43] are the BCD networks. EA [49], MTA [13], LRS [50], and our TIPA are the attack methods. (a) Results on the WHU-BCD dataset. (b) Results on the LEVIR-CD dataset.

being 12 × 12 pixels on the WHU-BCD dataset and 7.2 × 7.2 pixels on the LEVIR-CD dataset (i.e., 3.6 × 3.6 m). A single NVIDIA RTX 3090 GPU is used for both training and testing.

2) Target and Surrogate Models: To simulate real-world considerations, we select the surrogate models and target models in chronological order. We consider four target models: FCCDN [51], ISNet [41], DMINet [42], and CF-GCN [43]. On the WHU-BCD dataset, the target models, in the absence of adversarial patches applied to the posttemporal images, exhibit mIoU values of 91.37%, 89.94%, 88.30%, and 84.39%, respectively, accompanied by $F1$-scores of 90.94%, 89.28%, 87.31%, and 82.35%. Similarly, on the LEVIR-CD dataset, the respective mIoU values are 86.92%, 89.93%, 90.21%, and 89.66%, with corresponding $F1$-scores of 85.88%, 89.47%, 89.79%, and 89.18%. CLNet [64], SNUNet [65], and BIT [40] are used as the surrogate models. All the parameters of the models are trained on the clean original training set of WHU-BCD. In addition, the parameters of BCD networks are frozen during training and testing adversarial patches. And ResNet34 [63] is adopted as the backbone of all the listed models.

3) Competing Methods: Due to the constraints of the strict black-box scenario, where feedback from the target model is not available and there are no attacks for BCD, we opted

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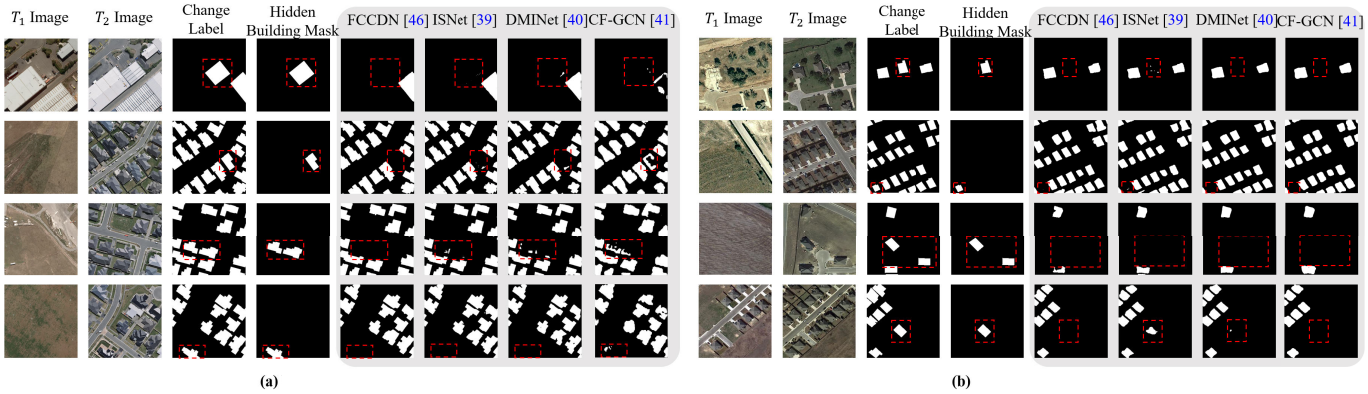


Fig. 10. Black-box visualization results with our TIPA adversarial patches on partial structures. FCCDN [51], ISNet [41], and DMINet [42] are the BCD networks. (a) WHU-BCD dataset. (b) LEVIR-CD dataset.

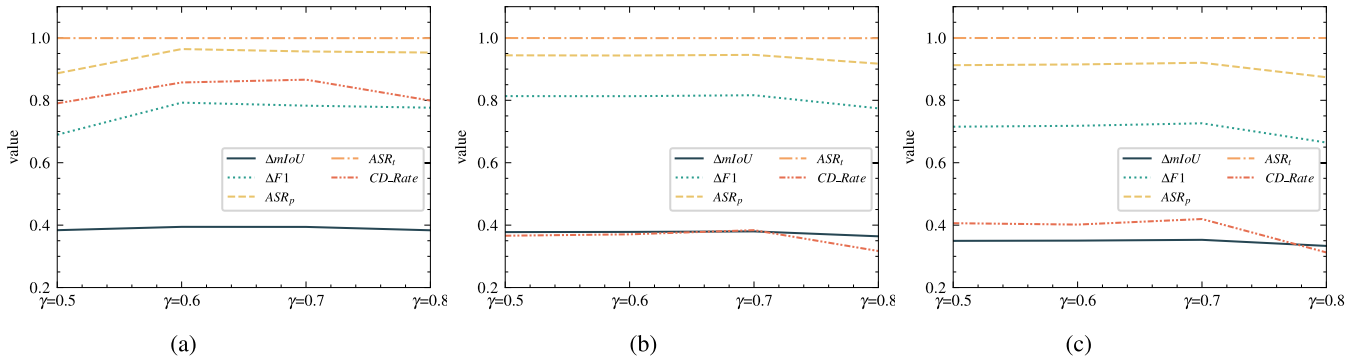


Fig. 11. Parameter analysis of γ on three target models. (a) Results on FCCDN. (b) Results on ISNet. (c) Results on DMINet.

for the plug-and-play components EA [49], MTA [13], and LRS [50] which are transfer-based black-box attacks as competing networks. EA is an ensemble attack, while MTA is a meta-thinking-based attack and LRS is an attack based on Lipschitz regularization. Note that MTA requires selecting a meta-surrogate model, and hence this method is experimented on CLNet, SNUNet, and BIT, respectively, and the average values are taken. CLNet, SNUNet, and BIT are chosen as the source models in each experiment of MTA. Similarly, LRS necessitates the selection of a surrogate model, and consequently, this approach is also trained on CLNet, SNUNet, and BIT, respectively, with average results being considered.

4) Experimental Results: Table III illustrates the results of several off-the-shelf black-box attack methods and our proposed TIPA on target models. It is evident that TIPA performs the best. Across all the buildings, TIPA not only surpasses the existing methods in attack capability but also enhances overall building concealment. Particularly, compared with the second best, CD_Rate shows improvements of 4.45%, 5.19%, and 3.34% on the three target models. FCCDN outperforms other target models due to the utilization of a nonlocal feature pyramid network that expands the receptive field, making it more susceptible to attack influences. It is worth mentioning that LRS, by applying Lipschitz regularization to the loss of the surrogate model, achieves a smoother and more controllable optimization process, facilitating the generation of more transferable adversarial examples. However, in our proposed FPW white-box adversarial sticker generation framework, the L_{TV} loss function already ensures smoothness. Consequently, when FPW is combined with the plug-and-play LRS for black-box adversarial sticker generation, the attack performance is less than satisfactory. Fig. 9(a) visualizes the results and trained adversarial patches of EA [49], MTA [13], LRS [50], and our method on the test set of WHU-BCD.

We also test the transferability of cross dataset. Table IV shows the result on the test set of LEVIR-CD, which the adversarial patch are trained on the training set of WHU-BCD. Compared with the second best, ASR_p improved by 1.38%, 9.29%, and 5.44% on the three target models, and CD_Rate improved by 0.75%, 5.07%, and 2.75% on the three target models. Fig. 9(b) visualizes the results of physical attacks on the test set of LEVIR-CD.

Upon visualizing the results, it is evident that our method typically hides more areas of the buildings. While some large buildings are completely hidden using MTA, the same buildings in our method results have only small regions unhidden, which have minimal impact on practical applications.

To demonstrate that change detection can proceed normally even in the presence of introduced perturbations, additional visualizations are conducted, with the results present in Fig. 10. From each dataset, four data samples are randomly selected, and perturbations are applied to some newly built buildings and not to others. The visualization results clearly indicate that change detection can accurately identify changes in undisturbed buildings and that only perturbed buildings are affected.

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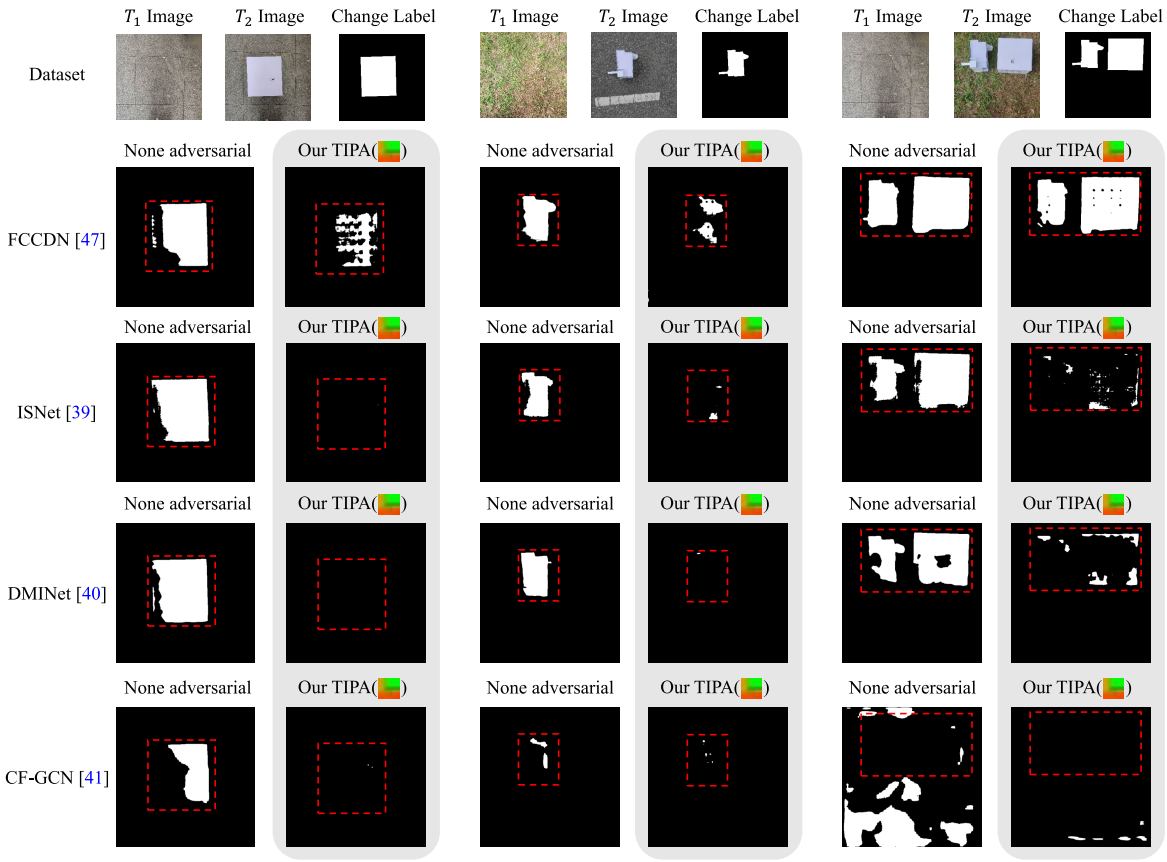


Fig. 12. Visualization results of our method in the real world. FCCDN [51], ISNet [41], DMINet [42], and CF-GCN [43] are the BCD networks.

The hyperparameter γ in (9) is analyzed experimentally. The results of three target models on the validation set of WHU-BCD are shown in Fig. 11. It can be observed from the graph that the model performances are not sensitive to hyperparameter γ, with the optimal performance achieved at $\gamma = 0.7$.

D. Experiments of Physical Attacks

1) Experimental Details: In the real-world scenarios, the target network of change detection is usually unknown, so physical attacks are tested on strict black-box scenarios. We test the patches which trained by our TIPA method. Specifically, we print and paste our adversarial patches on the 1:200 scaled Smith House and Gropius House and place them under different backgrounds and lighting conditions. Three scenes are selected to simulate the common environments in building changes, including bare land, grass, and residential areas, and filming is conducted from different directions and under different weather conditions. We collect a total of 100 sets of data using a Huawei P50 Pro phone. Each set of data includes bitemporal images and a posttemporal image with patches placed on new buildings.

2) Experimental Results: FCCDN [51], ISNet [41], DMINet [42], and CF-GCN [43] achieve results of 90.87%, 86.60%, 88.27%, and 62.76% on the clean original dataset of mIoU, respectively, with corresponding $F1$-scores of 91.40%, 86.82%, 88.65%, and 53.00%. The results of our physical attacks in the real world on target models are presented in Table V, which shows that our adversarial patches can successfully hide the new buildings even in the real world. And

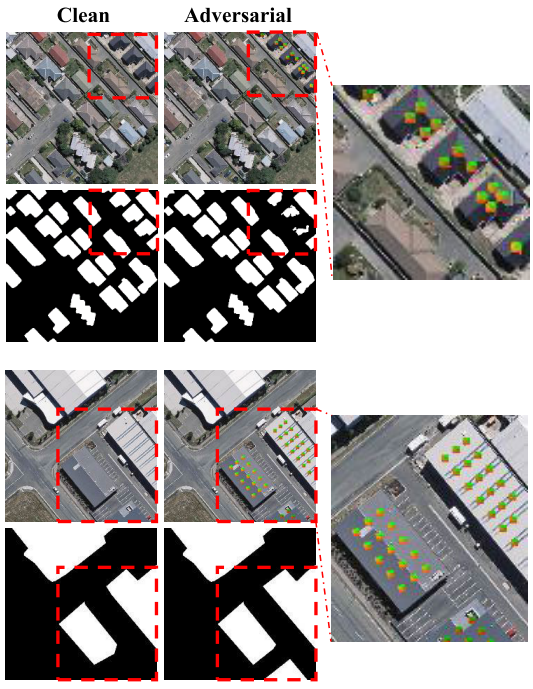


Fig. 13. Visualization results of posttemporal images for semantic segmentation.

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TABLE V
EVALUATION OF OUR TIPA METHOD IN THE REAL WORLD

BCD Networks	$\Delta mIoU$	$\Delta F1$	ASR_p	ASR_t	CD_Rate
FCCDN [51]	5.80	6.41	19.61	98.86	3.82
ISNet [41]	30.19	49.23	71.98	98.95	8.40
DMINet [42]	36.48	62.95	83.90	99.13	6.11
CF-GCN [43]	18.56	39.30	78.21	89.36	41.22

Fig. 12 shows the visualization results of the clean original images and the attack effects of our adversarial patches on target BCD networks.

V. DISCUSSION AND CONCLUSION

The existing researches lack the adversarial attacks for pixel-level tasks, thus this article focuses on physical attacks for BCD. We present the GPA framework to achieve generalized adversarial patches in the white-box scenario and the black-box scenario. Extensive experiments validate the effectiveness of our proposed methods in the digital and physical domains. But there is still room for improvement in fully concealing buildings, especially large ones. Moreover, conducting physical experiments on scaled-down models harbors inherent limitations, such as the material discrepancies between actual buildings and their scaled-down counterparts, as well as the complexity and diversity of real-world environments that surpass what can be simulated in scaled-down experiments.

The practical implications of our work include its application in hiding sensitive buildings, such as military bases on battlefields, which demonstrates its value in enhancing operational security. Furthermore, our method challenges the adversarial robustness of the existing change detection models, thus contributing to the advancement of adversarial defense in change detection.

We additionally conduct cross-task generalization experiments on another pixel-level semantic segmentation task. We select the widely used UPerNet model [66] using a pretrained Swin-L [67] as the backbone and trained on the training set of the WHU Building dataset [53]. A few images from the posttemporal images of the digital dataset are selected for testing, and the visualization results with our TIPA adversarial patches are shown in Fig. 13. It can be seen from the results that while some small buildings can be partially or completely hidden by the adversarial patches, medium or large buildings are not well hidden.

REFERENCES

[1] Y. Li, X. Li, W. Chen, and Y. Zhang, "A disparity-aware Siamese network for building change detection in bi-temporal remote sensing images," *Photogramm. Rec.*, vol. 39, no. 187, pp. 528–548, Sep. 2024. [Online]. Available: https://api.semanticscholar.org/CorpusID:269217302

[2] H. Li et al., "Using land long-term data records to map land cover changes in China over 1981–2010," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 4, pp. 1372–1389, Apr. 2017.

[3] L. Matikainen, J. Hyyppä, E. Ahokas, L. Markelin, and H. Kaartinen, "Automatic detection of buildings and changes in buildings for updating of maps," *Remote Sens.*, vol. 2, no. 5, pp. 1217–1248, Apr. 2010. [Online]. Available: https://www.mdpi.com/2072-4292/2/5/1217

[4] S. T. K. Jan, J. C. E. Messou, Y.-C. Lin, J. Huang, and G. Wang, "Connecting the digital and physical world: Improving the robustness of adversarial attacks," in *Proc. 33rd AAAI Conf. Artif. Intell. 31st Innov. Appl. Artif. Intell. Conf. 9th AAAI Symp. Educ. Adv. Artif. Intell.*, vol. 33, Jul. 2019, pp. 962–969, doi: 10.1609/aaai.v33i01.3301962.

[5] S. Liu, P. Y. Chen, X. Chen, and M. Hong, "SignSGD via zeroth-order Oracle," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2019, pp. 1–24. [Online]. Available: https://api.semanticscholar.org/CorpusID:108298677

[6] J. Chen and Q. Gu, "RayS: A ray searching method for hard-label adversarial attack," in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2020, pp. 1739–1747. [Online]. Available: https://api.semanticscholar.org/CorpusID:219981888

[7] L. Zhang and H. Fan, "Visual object tracking: Progress, challenge, and future," *Innovation*, vol. 4, no. 2, Mar. 2023, Art. no. 100402. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2666675823000309

[8] Q. Wang et al., "How to prevent malicious use of intelligent unmanned swarms?" *Innovation*, vol. 4, no. 2, Mar. 2023, Art. no. 100396. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2666675823000243

[9] A. Du et al., "Physical adversarial attacks on an aerial imagery object detector," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 3798–3808.

[10] Z. Yuan, J. Zhang, Y. Jia, C. Tan, T. Xue, and S. Shan, "Meta gradient adversarial attack," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7728–7737.

[11] F. Yin et al., "Generalizable black-box adversarial attack with meta learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 3, pp. 1804–1818, Mar. 2024.

[12] F. Yang et al., "Towards robust person re-identification by defending against universal attackers," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 5218–5235, Apr. 2023.

[13] Y. Qin, Y. Xiong, J. Yi, and C. Hsieh, "Training meta-surrogate model for transferable adversarial attack," in *Proc. 37th AAAI Conf. Artif. Intell. 35th Conf. Innov. Appl. Artif. Intell. 13th Symp. Educ. Adv. Artif. Intell.*, vol. 37, Jun. 2023, pp. 9516–9524, doi: 10.1609/aaai.v37i8.26139.

[14] Z. Huang and T. Zhang, "Black-box adversarial attack with transferable model-based embedding," in *Proc. 8th Int. Conf. Learn. Represent. (ICLR)*, Addis Ababa, Ethiopia, Jan. 2019, pp. 1–20. [Online]. Available: https://openreview.net/forum?id=SJxhNTNYwB

[15] H. M. Dolatabadi, S. M. Erfani, and C. Leckie, "AdvFlow: Inconspicuous black-box adversarial attacks using normalizing flows," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst.* Red Hook, NY, USA: Curran Associates Inc., Jan. 2020, pp. 15871–15884.

[16] Y. Li, J. Luo, Y. Zhang, Y. Tan, J.-G. Yu, and S. Bai, "Learning to holistically detect bridges from large-size VHR remote sensing imagery," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 11507–11523, Dec. 2024.

[17] Y. Li et al., "STAR: A first-ever dataset and a large-scale benchmark for scene graph generation in large-size satellite imagery," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 3, pp. 1832–1849, Mar. 2025, doi: 10.1109/TPAMI.2024.3508072.

[18] Y. Xu, B. Du, and L. Zhang, "Assessing the threat of adversarial examples on deep neural networks for remote sensing scene classification: Attacks and defenses," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1604–1617, Feb. 2021.

[19] C. Du, C. Huo, L. Zhang, B. Chen, and Y. Yuan, "Fast C&W: A fast adversarial attack algorithm to fool SAR target recognition with deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[20] C. Shi, Y. Dang, L. Fang, Z. Lv, and M. Zhao, "Hyperspectral image classification with adversarial attack," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[21] C. Shi et al., "Multifeature collaborative adversarial attack in multimodal remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5631815.

[22] Q. Hu, Z. Shen, Z. Sha, and W. Tan, "Multiloss adversarial attacks for multimodal remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4600813.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:27:33 UTC from IEEE Xplore. Restrictions apply.

[23] T. Bai, Y. Cao, Y. Xu, and B. Wen, "Stealthy adversarial examples for semantic segmentation in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614817.

[24] R. den Hollander et al., "Adversarial patch camouflage against aerial detection," *Proc. SPIE*, vol. 11543, pp. 77–86, Sep. 2020.

[25] M. Lu, Q. Li, L. Chen, and H. Li, "Scale-adaptive adversarial patch attack for remote sensing image aircraft detection," *Remote Sens.*, vol. 13, no. 20, p. 4078, Oct. 2021.

[26] Y. Zhang et al., "Adversarial patch attack on multi-scale object detection for UAV remote sensing images," *Remote Sens.*, vol. 14, no. 21, p. 5298, Oct. 2022. [Online]. Available: https://www.mdpi.com/2072-4292/14/21/5298

[27] C. Sun, J. Sun, X. Zhang, Y. Li, Q. Bai, and H. Sun, "Physical strip attack for object detection in optical remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5639611.

[28] D. Strong and T. Chan, "Edge-preserving and scale-dependent properties of total variation regularization," *Inverse Problems*, vol. 19, no. 6, pp. S165–S187, Nov. 2003, doi: 10.1088/0266-5611/19/6/059.

[29] M. Sharif, S. Bhagavatula, L. Bauer, and M. K. Reiter, "Accessorize to a crime: Real and stealthy attacks on state-of-the-art face recognition," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, New York, NY, USA, Oct. 2016, pp. 1528–1540, doi: 10.1145/2976749.2978392.

[30] L. Ding et al., "Towards universal physical attacks on single object tracking," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, no. 2, 2021, pp. 1236–1245, doi: 10.1609/aaai.v35i2.16211.

[31] Y.-C.-T. Hu, J.-C. Chen, B.-H. Kung, K.-L. Hua, and D. S. Tan, "Naturalistic physical adversarial patch for object detectors," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7828–7837.

[32] W. Cai, X. Di, X. Wang, W. Gao, and H. Jia, "Stealthy vehicle adversarial camouflage texture generation based on neural style transfer," *Entropy*, vol. 26, no. 11, p. 903, Oct. 2024. [Online]. Available: https://www.mdpi.com/1099-4300/26/11/903

[33] Y. Zhang, H. Foroosh, P. David, and B. Gong, "CAMOU: Learning physical vehicle camouflages to adversarially attack detectors in the wild," in *Proc. 7th Int. Conf. Learn. Represent. (ICLR)*, New Orleans, LA, USA, Sep. 2018, pp. 1–20. [Online]. Available: https://openreview.net/forum?id=SJgEl3A5tm

[34] Z. Xiao et al., "Improving transferability of adversarial patches on face recognition with generative models," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 11845–11854.

[35] W. Jia et al., "Fooling decision-based black-box automotive vision perception systems in physical world," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 7, pp. 7081–7092, Jul. 2024.

[36] W. Feng, B. Wu, T. Zhang, Y. Zhang, and Y. Zhang, "Meta-attack: Class-agnostic and model-agnostic physical adversarial attack," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7767–7776.

[37] S.-M. Moosavi-Dezfooli, A. Fawzi, O. Fawzi, and P. Frossard, "Universal adversarial perturbations," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 86–94.

[38] S. Thys, W. V. Ranst, and T. Goedemé, "Fooling automated surveillance cameras: Adversarial patches to attack person detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2019, pp. 49–55.

[39] K. Eykholt et al., "Robust physical-world attacks on deep learning visual classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1625–1634.

[40] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[41] G. Cheng, G. Wang, and J. Han, "ISNet: Towards improving separability for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5623811.

[42] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[43] W. Wang, C. Liu, G. Liu, and X. Wang, "CF-GCN: Graph convolutional network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5607013.

[44] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[45] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. 34th Int. Conf. Mach. Learn.*, vol. 70, Aug. 2017, pp. 1126–1135.

[46] Q. Wang, Y. Lv, Y. Feng, Z. Xie, and J. Huang, "A simple yet effective strategy to robustify the meta learning paradigm," in *Proc. 37th Int. Conf. Neural Inf. Process. Syst.* Red Hook, NY, USA: Curran Associates Inc., Jan. 2023, pp. 12897–12928.

[47] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst. (NIPS)*. Red Hook, NY, USA: Curran Associates, 2017, pp. 4080–4090.

[48] A. Raghu, M. Raghu, S. Bengio, and O. Vinyals, "Rapid learning or feature reuse? Towards understanding the effectiveness of MAML," in *Proc. 8th Int. Conf. Learn. Represent. (ICLR)*, Addis Ababa, Ethiopia, Jan. 2019, pp. 1–21. [Online]. Available: https://openreview.net/forum?id=rkgMkCEtPB

[49] X. Zhu, Z. Hu, S. Huang, J. Li, and X. Hu, "Infrared invisible clothing: Hiding from infrared detectors at multiple angles in real world," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13307–13316.

[50] T. Wu, T. Luo, and D. C. Wunsch, "LRS: Enhancing adversarial transferability through Lipschitz regularized surrogate," in *Proc. 38th AAAI Conf. Artif. Intell.*, vol. 38, Mar. 2024, pp. 6135–6143.

[51] P. Chen, B. Zhang, D. Hong, Z. Chen, X. Yang, and B. Li, "FCCDN: Feature constraint network for VHR image change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 187, pp. 101–119, May 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271622000636

[52] S. Flennerhag, A. A. Rusu, R. Pascanu, F. Visin, H. Yin, and R. Hadsell, "Meta-learning with warped gradient descent," in *Proc. 8th Int. Conf. Learn. Represent. (ICLR)*, Addis Ababa, Ethiopia, Apr. 2020, pp. 1–28. [Online]. Available: https://openreview.net/forum?id=rkeiQlBFPB

[53] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[54] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/10/1662

[55] J. Zheng, C. Lin, J. Sun, Z. Zhao, Q. Li, and C. Shen, "Physical 3D adversarial attacks against monocular depth estimation in autonomous driving," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 24452–24461.

[56] D. Wang et al., "FCA: Learning a 3D full-coverage vehicle camouflage for multi-view physical adversarial attack," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, 2022, pp. 2414–2422. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/20141

[57] K. K. Nakka and M. Salzmann, "Indirect local attacks for context-aware semantic segmentation networks," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Berlin, Germany: Springer-Verlag, 2020, pp. 611–628, doi: 10.1007/978-3-030-58558-7_36.

[58] Q. Shen, J. Huang, M. Wang, S. Tao, R. Yang, and X. Zhang, "Semantic feature-constrained multitask Siamese network for building change detection in high-spatial-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 189, pp. 78–94, Jul. 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271622001344

[59] C. Zhang et al., "A domain adaptation neural network for change detection with heterogeneous optical and SAR remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 109, May 2022, Art. no. 102769. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0303243422000952

[60] Y. Li et al., "MFVNet: A deep adaptive fusion network with multiple field-of-views for remote sensing image semantic segmentation," *Sci. China Inf. Sci.*, vol. 66, no. 4, Apr. 2023, Art. no. 140305.

[61] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Represent. (ICLR)*, San Diego, CA, USA, Bengio and Y. LeCun, Eds., Dec. 2014, pp. 1–15.

[62] Y. Li et al., "Meet: A million-scale dataset for fine-grained geospatial scene classification with zoom-free remote sensing imagery," *IEEE/CAA J. Autom. Sinica*, vol. 5, no. 12, pp. 1–20, May 2025.

[63] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[64] Z. Zheng, Y. Wan, Y. Zhang, S. Xiang, D. Peng, and B. Zhang, "CLNet: Cross-layer convolutional neural network for change detection in optical remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 247–267, May 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S092427162100068X

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[65] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[66] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Computer Vision—ECCV 2018*, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds., Cham, Switzerland: Springer, 2018, pp. 432–448.

[67] X. Guo et al., "SkySense: A multi-modal remote sensing foundation model towards universal interpretation for earth observation imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27662–27673.



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